Section 16.2: Line Integrals

Vector Calculus

Line Integrals: Integrating Along Curves

A **line integral** extends the definite integral from a straight interval $[a, b]$ to a **curve** C in 2D or 3D. Two types:

- **Scalar**: $\int_C f\, ds$ — add up a function weighted by arc length (e.g., mass of a wire)
- **Vector**: $\int_C \vec{F} \cdot d\vec{r}$ — add up a vector field along a curve (e.g., work done by a force)

The key ingredient from Chapter 13: the **arc length element**

$$ds = |\vec{r}\,'(t)|\, dt$$

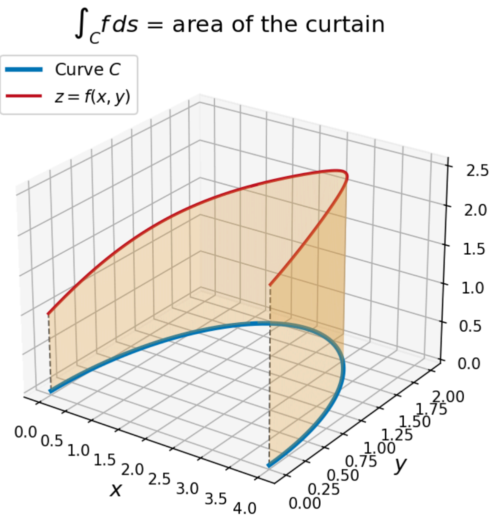
converts everything to an ordinary integral over $[a, b]$.

Line Integral of a Scalar Function

Line Integral with Respect to Arc Length

$$\int_C f\, ds = \int_a^b f\big(\vec{r}(t)\big)\, |\vec{r}\,'(t)|\, dt$$

Geometric interpretation: For $f > 0$, $\int_C f\, ds$ is the area of the "curtain" standing on C reaching up to $z = f(x, y)$.



The value does **not** depend on the parameterization — only on C and f. (When C is a segment of the x-axis, this reduces to $\int_a^b f(x)\, dx$.)

Recipe: (1) parameterize C as $\vec{r}(t)$, (2) compute $|\vec{r}\,'(t)|$, (3) substitute and integrate.

Example 1: Scalar Line Integral Along a Line Segment

Example 1

Evaluate $\displaystyle\int_C x \sin y \, ds$, where C is the line segment from $(0, 3)$ to $(4, 6)$.

Parameterize: $\vec{r}(t) = (1-t)\vec{r}_0 + t\vec{r}_1$ for $t \in [0, 1]$, then compute $|\vec{r}\,'(t)|$ and set up the integral.

Line Integrals with Respect to x and y

Besides integrating with respect to arc length (ds), we can integrate with respect to x or y individually:

Line Integrals with Respect to x and y

$$\int_C f(x, y)\, dx = \int_a^b f\big(x(t), y(t)\big)\, x'(t)\, dt$$

$$\int_C f(x, y)\, dy = \int_a^b f\big(x(t), y(t)\big)\, y'(t)\, dt$$

Key difference from ds: These use $x'(t)\, dt$ or $y'(t)\, dt$ instead of $|\vec{r}\,'(t)|\, dt$.

- ds always involves $|\vec{r}\,'(t)|$ — a magnitude, always positive
- dx and dy can be **negative** (they track direction)

These often appear **together** in the combined form:

$$\int_C P(x, y)\, dx + Q(x, y)\, dy = \int_C P\, dx + \int_C Q\, dy$$

This notation will become very important in Green's Theorem (Section 16.4).

$\int_C f\, ds$ uses the speed $|\vec{r}\,'(t)|$ and ignores direction. $\int_C f\, dx$ and $\int_C f\, dy$ track direction through the individual components $x'(t)$ and $y'(t)$.

Example 2: Line Integral Over a Piecewise Path

Example 2

Evaluate $\int_C x^2\,dx + y^2\,dy$, where C consists of the arc of the circle $x^2 + y^2 = 4$ from $(2, 0)$ to $(0, 2)$, followed by the line segment from $(0, 2)$ to $(4, 3)$.

Split C into C_1 (circular arc) and C_2 (line segment). Parameterize each piece separately, then add the integrals.

Example 3: Line Integral in 3D

Example 3

Evaluate $\int_C y\,dx + z\,dy + x\,dz$, where C is the line segment from $(1, 2, 3)$ to $(-1, 4, 6)$.

Parameterize: start at $\langle 1, 2, 3\rangle$, direction $= \langle -1, 4, 6\rangle - \langle 1, 2, 3\rangle = \langle -2, 2, 3\rangle$.

Line Integral of a Vector Field (Work)

For a varying force along a curved path, **work** = $\int_C (\text{force along path}) \cdot ds$.

Line Integral of a Vector Field

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}\big(\vec{r}(t)\big) \cdot \vec{r}\,'(t)\, dt$$

Three equivalent notations:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \vec{T}\, ds = \int_C P\, dx + Q\, dy + R\, dz$$

The connection: $d\vec{r} = \vec{r}\,'(t)\, dt = \langle dx, dy, dz \rangle$, so $\vec{F} \cdot d\vec{r} = P\, dx + Q\, dy + R\, dz$.

Quick Check: If $\vec{F} = \langle 1+y, x \rangle$, then $\int_C \vec{F} \cdot d\vec{r} = \int_C (1+y)\, dx + x\, dy$.

To evaluate: parameterize C, compute $\vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)$, and integrate from a to b.

Example 4: Work Done by a Force

Example 4

Let $\vec{F} = \langle 1+y,\ x \rangle$ and let C be the upper semicircle from $(-1, 0)$ to $(1, 0)$. Compute the work $\displaystyle\int_C \vec{F} \cdot d\vec{r}$.

Parameterize the upper semicircle from $(-1,0)$ to $(1,0)$. Hint: try $\vec{r}(t) = \langle -\cos t, \sin t \rangle$.

Properties of Line Integrals

Linearity:

$$\int_C (c_1\vec{F}_1 + c_2\vec{F}_2)\cdot d\vec{r} = c_1\int_C \vec{F}_1\cdot d\vec{r} + c_2\int_C \vec{F}_2\cdot d\vec{r}$$

Piecewise paths: $C = C_1 \cup C_2 \cup \cdots \;\Rightarrow\; \int_C = \int_{C_1} + \int_{C_2} + \cdots$ (used in Example 2)

Reversing direction:

$$\int_{-C} \vec{F}\cdot d\vec{r} = -\int_C \vec{F}\cdot d\vec{r} \qquad \text{but} \qquad \int_{-C} f\,ds = \int_C f\,ds$$

Vector line integrals **change sign**; scalar line integrals **don't** (arc length has no direction).

> Reversing direction negates $\int_C \vec{F}\cdot d\vec{r}$ because $d\vec{r}$ reverses, but doesn't change $\int_C f\,ds$ because ds is always positive.

Homework

Section 16.2: 1, 3, 5, 7, 11, 13, 19, 21, 23, 35, 43, 45, 47, 49